



**MASTERMIND
INDIA**

Spirit of India Special

Virendra KUMAR SINHA, 58

INNOVATION

**Generator accessory
for cleaning exhaust**

COST OF PRODUCTION

Rs 2 lakh

WHAT IT DOES

**It controls pollution by
removing carbon from
exhaust gases and
muffles the sound**

CARBON CREDIT

A school dropout, Virendra Kumar Sinha could not have imagined that his childhood fetish for repairing almost anything—from sewing machines to air pumps to bicycles—would yield a pollution control device for generators. Sinha's device precipitates suspended particulates present in exhaust gases, making it clear and reducing its temperature. The first of its kind, this attachment can be commercially used in generators and engines of any kind. A self-taught welder and fabricator from Motihari in Bihar, Sinha has priced the attachment at Rs 2 lakh. He gives the credit for his success to a supportive family, particularly his wife Sudha. "That apart, hard work and honesty are everything. Success eventually comes if there is determination," he says.

For the moment, Sinha is happy clearing the air over Motihari, where the idea germinated. After a stint as a road and building contractor in the

1970s, he set up his own fabrication unit for manufacturing iron gates and grill works. Limited resources forced him to work out of a cramped space adjacent to his house in a congested colony and use a diesel generator. "The noisy genset and smoke emanating from it disturbed children in a nearby school and angered the residents," Sinha recalls. It was an ultimatum served by a court to either find a solution or shut shop that made Sinha design the attachment for minimising the genset's noise and smoke.

From 2000 to '06, Sinha experimented with various models and after several trials developed a chamber to muffle the sound and break down the exhaust components using an iron drum, perforated iron pipes, steel channels, baffles and a soot collecting tray. Soon Sinha created the first prototype. Further improvisations and Rs 2 lakh of investment finally resulted in the genset accessory that met needed standards. "It cut the sound level such

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SINHA AT HIS
WORKSHOP

THE DEVICE CLEANS EXHAUST FUMES BY FILTERING OUT THE SUSPENDED PARTICULATES.

that neighbours began to enquire if all was well in the workshop." After 3,000 hours of engine operation, the device removes about 5 kg of soot and collects 12 to 14 kg of carbon. It has drawn a favourable response from industry. Sinha also has a patent and has been honoured by the Government. He is now working to fine-tune it for advanced applications. "It started from my need to survive. It has become an issue of prestige now," he says.

By Nivedita Mukherjee